

AI Jobs Salary Prediction (2025–2026) – Simple Explanation Notes

Project Title

AI Jobs Salary Prediction (2025–2026)

Problem Type

Supervised Learning – because the dataset contains input features and a known output (salary).

Machine Learning Category

Regression – the model predicts a continuous numeric value such as salary.

Model Used

Random Forest Regressor – an ensemble learning algorithm that combines many decision trees to improve prediction accuracy.

How the Algorithm Works

- Creates multiple decision trees.
- Each tree learns patterns from the dataset.
- Each tree makes a salary prediction.
- The final prediction is the average of all tree predictions.

Features Used

- Job Role
- Country
- Company Size
- Industry
- Remote Work Type

Evaluation Metrics

- MAE (Mean Absolute Error): Average difference between predicted and real salary.
- MSE (Mean Squared Error): Squares the errors to penalize larger mistakes.

- R^2 Score: Shows how well the model explains the data (1.0 is perfect).